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HVAC SYSTEMS & EQUIPMENT INSTALLATION

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TRANSMISSION CONSTRUCTION STANDARD

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1.0 SCOPE

1.1 This Transmission Construction Standard (TCS) prescribes the minimum mandatory requirements governing the fabrication, installation and testing of Heating, Ventilating and Air Conditioning (HVAC) and Refrigeration Systems for facilities (i.e., building offices, control rooms, substations, communications, etc.) in National Grid SA Transmission network.

1.2 Conflicts between this standard and other Engineering Standards, related Material Specification, Forms and Standard Drawings shall be resolved by the Manager, Standards and Specifications Department.

1.3 Any deviations providing less than the minimum requirements of this standard require written approval from the Standards and Specifications Department Manager.

2.0 REFERENCE STANDARDS

AABC	Associated Air Balance Council, National Standards for Total System Balance
AMCA	Air Movement and Control Association
ANSI B31.9	Building Services Piping
ANSI/ASHRAE STD. 15	Safety Standard for Refrigeration Systems
ASHRAE Guideline 1	HVAC & R Technical Requirements for the Commissioning Process
ASHRAE STD. 111	Testing, Adjusting, and Balancing of Building HVAC Systems.
ASHRAE Handbook	HVAC System & Equipment
ASHRAE Handbook	HVAC Application
ASHRAE STD 52	Method of Testing Air-Cleaning Devices Used In General Ventilation For Removing Particulate Matter
ASME B31.9	Building Services Piping
ASTM	American Society for Testing and Materials
AWS D1.1	Structural Welding Code - Steel

NEBB	National Environmental Balancing Bureau, Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems
NEMA	National Electrical Manufacturers Association
NFPA 70	National Fire Protection Association: National Electrical Code
NFPA 90A	Installation of Air Conditioning and Ventilating Systems
NFPA 101	Life Safety Code.
SMACNA	Ducted Electric Heat Guide for Air Handling Systems
SMACNA	HVAC Systems - Testing, Balancing and Adjusting
SMACNA	Accepted Industry Practice for Industrial Duct Construction
SMACNA	Round Industrial Duct Construction Standards
SMACNA	Fibrous Glass Duct Construction Standards
SMACNA	Rectangular Industrial Duct Construction Standards
SMACNA	HVAC Air Duct Leakage Test Manual
SMACNA	HVAC Duct Construction Standards Metal and Flexible
SMACNA	Fire, Smoke and Radiation Damper Installation Guide for HVAC Systems
SMACNA	Installation Standards for Residential Heating and Air Conditioning Systems
BS EN 13053	Ventilation for buildings - Air handling units - Ratings and performance for units, components and sections
BS EN 1886	Ventilation for Buildings - Air Handling Units - Mechanical Performance
UMC	Uniform Mechanical Code
UL-181	Duct Rated Class I

3.0 GENERAL REQUIREMENTS

3.1 All equipment shall be installed in accordance with manufacturer's recommendations but in all cases in compliance with the requirement set forth in this Standard and local governing codes.

3.2 It is recommended that indoor equipment shall be located as centrally as practical and within the insulated envelope of the building to minimize capacity losses.

3.3 Indoor equipment shall be installed in such location and manner that can be easily removed or serviced without major alterations to structure, attached ductwork or piping.

3.4 Indoor equipment and systems shall be installed to provide isolation from direct contact with supply and return air ducts. Flexible connections shall be installed between the equipment and duct.

3.5 Outdoor equipment shall be located to avoid arrangements that may cause restriction, or recirculation of air flow. Also, it should be located as far as practically away from working areas to prevent any noise radiated by the equipment.

3.6 Any air handlers, refrigeration condenser units, pumps, or any equipment mounted outside, shall be mounted on a concrete housekeeping pad, at a minimum of 100 mm above the floor or 150 mm above the surrounding ground finished grade level. Care shall be exercised to eliminate potential vibration carry-over to the structure.

3.7 Roof mounting of HVAC equipment can be considered if there is space restriction within the building area provided that safe access to roof top is facilitated.

3.8 Access doors shall be provided to every valve, damper or mechanical device that requires periodic inspection or adjustment. Access doors for fire/smoke dampers shall be positioned so that dampers can be reset and fusible link replaced, and periodically tested.

Electric duct heater shall also be provided with access door for periodic inspection and maintenance.

3.9 Provide a minimum of 900 mm clearance on each side of air handling unit with openable panel or controls, unless a greater clearance is required by the manufacturer. Adequate space shall be provided for coil removal.

3.10 No work shall be covered or otherwise concealed without the Company inspection. Any duct work, piping, or components so covered without being tested and inspected shall be uncovered for inspection.

3.11 All HVAC equipment, ductwork and piping shall be coordinated with all other trades (Structural, Electrical, Architectural, Fire Protection, etc.) to determine conflicts or interference.

Route piping (hydronic and refrigerant) in orderly manner, parallel to building structure and maintain gradient. Sleeve pipe passing through partitions, walls and floors.

Any interference between ducts, piping, equipment, shall be resolved by the concerend departments.

3.12 Each equipment unit shall be installed at the location indicated on the drawings and in accordance with details of mounting, with vibration isolation as required by the Design Specifications.

3.13 All equipment shall be properly grounded in accordance with NEC Art. 250 and TES-P-119.10 - Grounding.

3.14 All system components, equipment and materials received at job site shall be checked for damages and shortages. Materials shall also be checked against packing lists for correctness of materials as to size, model and quantity. The crate shall be re-nailed or containers resealed after inspection has been made to insure protection of equipment prior to installation.

3.15 Suitable storage facilities shall be provided to protect system components and materials from weather and pilferage. Whenever possible, the materials or equipment should not be permanently removed from boxes or crates until such time equipment is to be installed.

3.16 HVAC equipment shall be preconditioned by a certified and competent refrigeration mechanic either on the job site or in the shop prior to installation in accordance with vendor's recommendations. A detailed check shall be made to see that all components are furnished. Attention to this point will avoid costly dismantling of the system after installation.

3.17 A manufacturer’s warranty as per National Grid SA requirements shall be provided under after substantial completion of project which shall cover complete assembly of equipment.

3.18 The HVAC control system shall be designed and operated to receive an input signal from Fire Alarm Control Panel to shutdown the HVAC system and to close the motorized fire/smoke dampers associated with the operation of the Extinguishing System as per TES-B-106.01,TES-K-100&SAF -01of HCIS local regulations.

3.19 These requirements shall be read in conjunction with relevant clauses of TES-P-119.19.

4.0 AIR SYSTEMS

4.1 General

4.1.1 Air System and ductwork shall be installed according to the Construction Drawing and Specification and Manufacturer's Installation Manual.

4.1.2 All air openings shall be furnished in accordance with drawings and specifications, and mounted at locations shown on drawings.

4.1.3 Ducts shall be installed in accordance with Uniform Mechanical Code (UMC).

4.2 Air Handling Units (Chilled Water Coil or Direct Expansion Coil)

4.2.1 Evaporators of air handling units shall be mounted a minimum of 150 mm off the floor, to allow room for condensate trap. A corrosion resistant drain pan shall be provided under cooling coil.

4.2.2 Neoprene pad or spring vibration isolators compatible for the air handling unit shall be provided so as not to transmit vibration in the building.

4.2.3 A trapped drain shall be attached to the bottom or from the bottom side of the cooling coil drain pan. This trap shall have a minimum of 50 mm trap height, with discharge piped to the nearest plumbing floor drain (with air gap), installed in accordance with ASHRAE Equipment Handbook. The drain line size shall not be less than the drain pan size connection. The trap shall have three cleanouts.

Units installed outside the building may discharge condensate into the ground, planters or seepage pits, but not into walkways.

4.2.4 Evaporator inlet and outlet connections shall be provided with dielectric union if evaporator coils and piping or tubing connected to it are made of dissimilar materials.

4.2.5 A strainer shall be provided upstream of the control valve at the chilled water inlet to the cooling coil and an isolation valve shall be provided.

4.2.6 The entering chilled water shall be connected to evaporator coil on air exit or leaving side for best heat transfer. This type of heat transfer arrangement is called counter flow.

4.3 Fans

4.3.1 Install fans level and plumb, in accordance with manufacturer's written instructions. Support units as described below, using the vibration control devices indicated.

a) Support floor-mounted units on concrete equipment bases using neoprene pads. Secure units to anchor bolts installed in concrete equipment base.

b) Support floor-mounted units on concrete equipment bases using housed spring. Secure units to anchor bolts installed in concrete equipment base.

c) Suspended Units: Suspend units from structural steel support frame using threaded steel rods and vibration isolation springs.

4.3.2 Arrange installation of units to provide access space around for service and maintenance.

4.3.3 Install back draft dampers on inlet to roof and wall exhausters.

4.3.4 Provide back draft dampers on outlet from cabinet-type and ceiling exhaust fans and as indicated.

4.3.5 Do not operate fans for any purpose until ductwork is clean, filters in place, bearing lubricated and fan has been test run under observation.

4.4 Ductwork, Air Devices and Accessories

4.4.1 General

a) Ductwork shall be fabricated, installed and sealed in accordance with latest accepted good practices for sheet metal work and in accordance with the HVAC Duct Construction Standards by Sheet Metal and Air Conditioning Contractors National Association (SMACNA) Manual, National Fire Protection Association (NFPA) 90A, and Project Specifications and the Drawings.

b) Prior to fabrication, the Contractor shall review drawings of all disciplines including Architectural, Structural, Mechanical, Electrical, Plumbing, etc., and make on-site measurements to ascertain that no interference will be encountered upon installation. If there are any significant deviations from the Contract Drawings, produce "Interference Drawings". Before fabricating the sheet metal work and installing related equipment, the Contractor shall send a letter stating that no interference exist in the proposed installation. The letter must be both signed by the Mechanical Sub-Contractor and the Main or General Contractor. By not submitting this letter, the Contractor certifies that he has performed the above requirements and no interference will result in installation.

c) Ductwork and appurtenances shall be fabricated in accordance with details as shown on the drawings. Provision shall be made in the fabrication of developed lengths over four feet of duct work to allow tolerances of one inch per two lineal feet of run to account for discrepancies in locations of turns and outlets.

d) All ductwork is to be inspected prior to assembly to be sure that all components which control air deflection have been installed according to the drawings. In addition, all pieces of ductwork shall be cleaned to remove all foreign matter such as wood, rags, and heavy concentrations of dust or sand.

e) Air system equipment and ductwork shall be installed according to the Construction Drawings and Specifications and Manufacturer's Installation Manual.

f) All air system supply and return grilles/regisres shall be furnished in accordance with drawings and specifications, and mounted at locations shown on drawings.

g) All duct openings shall be properly covered to avoid dust intrusion until final connections to the terminal devices (supply outlets and return inlets) or adjacent ductwork are to be connected. The cover shall be made of material suitable for leak test of ductwork.

h) Ductwork shall never be hanged or supported from adjacent pipes, electrical conduits and other ductwork.

i) All duct joints shall be applied with approved duct sealant. Leaks in duct joints shall be corrected by applying the same duct sealant.

j) Flexible ducting without insulation is not acceptable in the air distribution system.

k) Branch duct and flexible ducting joint or connection shall be sealed with duct tape and then clamped.

l) Duct sizes shall be inside clear dimensions. For lined ducts, when specified, maintain sizes inside lining.

m) Use double nuts and lock washers on threaded rod supports.

n) During Construction, provide temporary closures of metal or taped polyethylene on open duct to prevent construction dust from entering ductwork system.

4.4.2 Duct Access and Inspection Provision

a) Service opening or removable duct section shall be provided adjacent to each fire door, fire damper, smoke damper, combination fire/smoke damper, smoke detector, duct heater, volume damper (motor operated and manual), turning vanes, splitter damper, etc. The opening shall be large enough to permit maintenance and resetting of the device.

b) The service openings or removable duct sections shall be identified with letters not less than 12.7mm in height to indicate and identify the location of the fire protection equipment.

4.4.3 Duct Insulation

a) Ducts shall be insulated with the specified insulation as indicated on the construction specifications and project drawings.

b) All quadrants and other balancing devices shall be left exposed for operation or inspection with the insulation brought as close to within 25mm of the balancing devices and then sealed at its edges.

c) Duct coverings shall not conceal any service opening. The insulation shall be brought as close to within 25mm of the balancing devices and then sealed at its edges.

d) Duct linings shall be interrupted at fire dampers and fire doors so as not to interfere with the operation of the devices.

e) Exposed duct insulation shall be protected from all weather and provided with covering that is resistant to ultra violet radiation.

f) Apply adhesive and coating at the manufacturer's recommended average application per liter (gallon) rate.

g) Keep insulation materials dry during application and finishing.

4.4.4 Terminal Devices, Air Outlets And Air Inlets

a) Terminal devices, air outlets and air inlets shall be furnished in accordance with the article described in the construction specifications or project drawings, and shall be checked before installation to see that the balancing devices and adjustable bars or cones function easily without force.

b) Outlets and inlets shall installed at the locations as shown on the drawings and in accordance with the details on the project drawings.

c) When located less than 2.13 m (7 ft) above the floor, inlet and outlet openings shall be protected by a substantial grill or screen having an opening through which a 1.27 cm (1/2 in.) sphere will not pass.

d) Avoid installing diffusers or grilles directly into the bottom or sides of the main air duct. No amount of adjustment will decrease the noise level generated.

e) Install air outlets and inlets to ductwork with airtight connection.

4.4.5 Fresh Air Intake

a) Bird screen made of corrosion resistant material, not larger than 13mm (1/2") mesh, shall be installed at fresh air intake.

b) Intakes shall be located to avoid drawing in combustible material or flammable vapor and to minimize the hazard from fires in other structures.

c) Flashing for fresh air intake shall be tight rain proof.

d) Electronic air cleaners or washable type filters shall be installed in the outdoor air intake ducts of building including sand trap and volume damper.

e) The outdoor air intakes must be located where there is the least possibility of drawing smoke back into the air conditioning and ventilating systems.

4.4.6 Filters

a) Air filters shall be installed ahead of cooling coils, heating coils and other air conditioning equipment in the system to protect them from dust.

b) Sufficient space shall be provided in front of or behind the filter, or both, depending on its type, to make it accessible for inspection and service. A distance of 500 mm to 1000 mm may be required, depending on the filter chosen.

c) Access doors of convenient size shall be provided to the filter service areas, and shall have lockable latches for easy opening during maintenance and inspection.

d) All doors on the clean air side should be gasketed to prevent infiltration of unclean air. Any filter bank must be caulked to prevent bypass of unfiltered air. This is most important when high efficiency filters are used.

e) Filters installed close to an air inlet (fresh air make-up) shall be protected from the weather by suitable louvers, in front of which a large mesh wire screen and sand trap shall be installed .

f) Filters, other than electronic air cleaners, shall have permanent indicators to give a warning when the filter resistance reaches too high a value or is exhausted, as with automatic roll media filters.

g) Electronic air cleaners shall have an indicator or alarm system to indicate when power supply is off or shorted out.

h) Install filter gauge static pressure tips upstream and down stream of filters. Mount filter gauges on outside of filter housing or filter plenum, in accessible position. Adjust and level.

i) Do not operate fan system until filters (temporary or permanent) are in place. Replace temporary filters used during construction and testing, with clean sets.

j) Washable type air filters shall be highly efficient made up of 1.5mm. thick extruded aluminum channel. Filter media shall be of multi layer aluminum wire screen and covered by expanded slit bonded layers of aluminum mat mesh.

4.4.7 Duct Heater

a) Where electrical resistance duct heaters are installed in the ducts, the installation shall be made so as to avoid creating a fire hazard. Ducts rated Class 1, UL 181, duct coverings and linings shall be interrupted at the immediate area of operation of such duct heaters in order to meet clearances specified as a condition of the equipment listing.

b) The installation of electrical duct heaters shall comply with the provisions of NFPA 70, National Electrical Code.

c) To ensure an even distribution of air across the heater surface, locate the heater at least 1.20 meters away from elbow, transition, filter or frame member or fan, unless a greater clearance is required by the manufacturer.

5.0 CHILLED WATER SYSTEM

5.1 Air Cooled, Packaged Water Chiller

5.1.1 When selecting a site for installation, consider the following guides:

a) For outdoor location of the unit, select a place having a minimum sun exposure and an adequate supply of fresh air for the condenser.

b) Avoid locations beneath windows or between structures where normal operating sounds may be objectionable.

5.1.2 The unit must be installed with sufficient clearances for air entrance to the condenser coil, for air discharge away from the condenser, and for servicing access. The minimum clearances shall be in accordance with the vendor's recommendations.

5.1.3 The unit shall be mounted on a flat and level foundation, ground or roof, capable of supporting the entire operating weight of the equipment and with vibration isolators. Make sure that all compressor inside the package chiller is mounted on a spring type vibration isolators.

5.1.4 For roof installation, choose a spot with adequate structural strength to safely support the entire weight of the unit and service personnel. Care must be taken not to damage the roof during installation. If the roof is bonded or water proofed, the building architect or Contractor shall be consulted for special installation requirements.

5.1.5 For ground installation, it is important that the units be installed on a substantial base or foundation that will not settle, causing strain on the piping system and resulting in possible leaks. Additionally, the equipment foundation shall not be tied to the main building foundation to minimize noise transmission. Precautions shall be taken to protect the unit from tampering by unauthorized persons.

5.2 HVAC Pumps

5.2.1 Pipe size shall not be smaller than pump nozzles and preferably pipe size shall be larger than the nozzle size on the suction side.

5.2.2 From a larger suction line to the pump nozzle, an eccentric reducer must be used, keeping the pipe flat on top to prevent formation of air pockets in the pipe.

5.2.3 Whenever possible, tapered reducers on the suction and increasers on the discharge should be installed directly on the pump nozzle and shall be suitably sized to ensure efficient flow on the system and conserve power.

5.2.4 Never place a pipe-line elbow in a horizontal plane directly at the pump suction nozzle.

5.2.5 Between the elbow and the suction nozzle, use a piece of straight pipe at least 4 to 6 pipe diameters long.

5.2.6 Suction piping shall be provided with vibration isolator or flexible connector, isolation valve, strainer with blow-off valve, and pressure gauge.

5.2.7 Discharge piping shall be provided with pressure gauge, check valve, isolation valve and flexible connector.

5.2.8 On the discharge line, the check valve shall be placed between the pump and the gate valve , and the tapered increaser between the pump and check valve.

5.2.9 Pressure gauges shall be installed as closed as possible to the pump suction and discharge. (Install gauges at integral pressure gauge tapings where provided).

5.2.10 The suction line approach to the pump shall be as straight as possible and all elbows shall have long radii.

5.2.11 A straight section of pipe shall be attached to the suction inlet to allow the fluid to straighten out before entering the pump; this is especially true of double suction pumps.

5.2.12 Suction line shall be at least one or two sizes larger than the pump suction nozzle. In any case the required Net Positive Suction Head (NPSH) shall decide the suction line size.

5.2.13 Suction line shall be airtight, with no high spots where air or gases may separate out of the fluid.

5.2.14 Both suction and discharge pipe connections must be supported separately and in such a way as to impose no strain on the pump.

5.2.15 Rubber-in-shear or corrugated rubber shall be used for pumps on ground floor installations. In more critical applications, on floors above occupied areas (especially those of executive offices, board rooms, libraries, hospital areas) steel spring isolation is recommended for pump installation.

5.2.16 Always use the discharge valve whenever the capacity of pump has to be controlled by throttling.

5.2.17 Check, align and certify alignment of base-mounted pumps prior to start-up.

5.2.18 Lubricate pump before start-up.

5.2.19 Chemical shot feed system shall be provided at pump suction and discharge piping.

5.2.20 Hydraulic Institute Standards shall also be consulted for installation of hydronic system.

5.3 Hydronic Piping

5.3.1 All piping shall be run as directly as possible with a minimum of elbows and other fittings.

5.3.2 The chilled water piping system shall be laid out so that the circulating pump discharges into the chiller or cooler.

5.3.3 Hand stop valves shall be provided where required to facilitate servicing. Drain connections shall be provided at all low points to permit complete drainage of the chiller and the system piping. Additionally, a strainer (40 mesh) is recommended for use on the inlet line to the chiller.

5.3.4 Thermometers and pressure gauges shall be provided in the inlet and outlet water lines of the chiller and AHU. Pressure gauges shall be glacier filled.

Pressure gauges shall be installed as closed as possible to the chilled water outlet and inlet of the chiller.

5.3.5 A flow switch shall be installed in the leaving water piping of the chiller.

5.3.6 Chilled water systems shall be forced re-circulating systems. Balancing valves shall be installed in discharge of each water coil downstream of the control valve. When 3-way valves are used, there shall be a balancing valve in the bypass line. Isolation valve shall be installed on water supply coil. Manual bypass valves shall be provided around control valves and flow meters.

5.3.7 Pete's plug (insertion tap points) or thermometer wells as required shall be installed on supply and return pipe branch to each chiller, water coil and on each side of control valves. Gauge cocks shall be provided on each side of pumps and flow meters.

5.3.8 To eliminate piping noise caused by free air, the system must be equipped with air separation device to minimize the amount of entrained air in the piping circuit. Air should be vented at the highest point of the system.

5.3.9 Provide access where valves and fittings are not exposed.

5.3.10 Install valves with stems upright or horizontal and not inverted.

5.3.11 Use unions and coupling downstream of valves and at equipment or apparatus connections. Do not use direct welded or threaded connections to valves, equipment or other apparatus.

5.3.12 Use non-conducting dielectric connections whenever jointing dissimilar metals.

5.3.13 Use graduated balancing valves or globe valves for throttling, bypass or manual flow control services.

5.3.14 Provide hangers and supports in accordance with ASME B31.9.

5.3.15 Welding materials and procedures shall be as per ASME Code.

5.3.16 Provide inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams . Provide hooked rod to concrete reinforcement section for inserts carrying pipe over 100 mm diameter.

5.3.17 Provide expansion tanks on suction side of the system circulating pumps to take care of expansion of liquid on system.

5.3.18 Provide relief valves on expansion valves. Select system relief valve capacity to exceed the rating of connected equipment.

6.0 REFRIGERATION SYSTEM

6.1 Refrigerant

6.1.1 Use only refrigerant in any equipment for which the equipment was designed by the Manufacturer. Use only one refrigerant in a system.

6.1.2 The refrigerant shall be delivered to the job in original containers.

6.2 Piping Materials

6.2.1 Unless otherwise specified, all refrigeration piping shall be refrigeration grade, Type K hard drawn degreased sealed copper tubing (ASTM B-280).

6.2.2 Extreme care shall be taken to keep all refrigerant piping clean and dry. It shall be kept sealed except when cutting or fabricating. Each length shall be inspected and swabbed with a cloth soaked in refrigeration oil if any dirt, filings, or visible moisture are present.

6.2.3 All sweat-type fittings shall be wrought copper (ANSI/ASME B16.22) or forged brass. All elbows and return bends shall be of the long radius type. If flare fittings are required, they shall be of the frost proof type, (except on connections not subject to condensation), and constructed of forged brass. Soldered joints are preferred and shall be used wherever practical.

6.3 Piping Installation

6.3.1 Tubing shall be installed in a neat and workman-like manner. All lines shall be supported at intervals of not more than 2.44 m and suitably anchored. Rubber grommets shall be used between tubing and clamps to prevent line chafing.

6.3.2 Where vertical risers of more than 1.53 m occur in a suction line, the riser shall be trapped at the bottom.

6.3.3 Where a branch suction line enters a main suction line it shall enter at the top. Piping shall be arranged so refrigerant or oil cannot drain from the suction line into the coil.

6.3.4 Individual fixture or unit suction and liquid lines shall be of the size recommended by the Manufacturer as shown in the applicable installation and service instructions manuals. Liquid and hot gas refrigerant lines shall be sized in accordance with good industry practice to avoid excessive pressure drops. Branch and main suction lines shall be sized to maintain adequate velocities to properly return oil to the compressor under minimum load conditions at the lowest saturated suction pressure to be expected.

6.3.5 All joints in the compressor discharge line shall be brazed with a suitable high temperature silver solder alloy containing not less than 15% silver. Use only a suitable silver solder alloy on all copper to copper connections in the suction line and liquid line. At any copper to brass joint where damage could occur from excess heat use 95/5 solder. Use a solder with at least 35% silver content on all copper to steel, brass to steel, or steel to steel joints. During the brazing operation, dry nitrogen must be bled through the piping at very low pressure 6.894 kPa (1.0 psig) to prevent oxidation and scaling.

6.3.6 To prevent contamination of the line internally, limit the soldering paste or flux to the minimum required. Flux shall be applied to the male portion of the connection, never the female.

6.3.7 Suction lines from low temperature cases shall be insulated where run below the floor level. All exposed suction lines, both low and medium temperature, shall be insulated as necessary to prevent condensation.

6.3.8 Insulation shall be of the closed cell type with weatherproof finish, shall fit the tubing snugly, and shall be applied and sealed in accordance with the Manufacturer's instructions. Minimum insulation wall thickness shall be 12.7 mm.

6.3.9 The refrigerant piping shall be adequately protected. Permanent guards shall be installed as required to protect the piping and fittings from damage. Metal pipe sleeves shall be provided where tubing passes through a concrete wall or floor, and the space around the tubing shall be filled with a master insulating compound.

6.3.10 Arrange the piping so that normal inspection and servicing of the compressor and other equipment is not hindered. Do not obstruct the view of the crankcase oil sight glass, or run piping so that it interferes with removal of the compressor or other components.

6.3.11 Tubing installed in trenches or conduit under the floor must be level to prevent oil trapping. Guard against deformation or damage from trucks carrying heavy loads, or cement being poured.

6.3.12 Refrigerant piping should be protected from contact with dissimilar metals to prevent corrosion.

6.3.13 Locate expansion valve sensing bulb immediately downstream of evaporator on suction line.

6.3.14 Install flexible connectors at right angles to axial movement of compressors, parallel to crankshaft to eliminate excessive vibration.

6.4 Installation of Accessories

6.4.1 Vibration eliminators shall be installed in the suction and discharge lines of all compressors with spring or flexible mounting. The vibration eliminator must be applied according to the Manufacturer's recommendations. For Copelametic compressors, the vibration eliminator should be mounted parallel to the crankshaft, as close to the compressor as possible. Installation in a horizontal plane at right angles to the crank- shaft is not acceptable, since the resulting stress from compressor movement may cause failure of the vibration absorber. If installed in a vertical position, the eliminator joints must be sealed against dripping from condensation to protect from freezing.

6.4.2 A solder type combination liquid sight glass and moisture indicator shall be installed in each system and located for easy visibility.

If liquid line dryers are not otherwise specified, they shall be of the replaceable drier core type, and of the size recommended by the Manufacturer. Drier cartridges shall not be installed until the second evacuation has been completed. Shut-off valves shall be installed in each inlet and outlet pipe connections.

6.4.3 Two evacuation fittings are necessary. One shall be in the suction line at the inlet side of the suction line filter, and one shall be in the liquid line at the outlet side of the filter- drier. If properly valved, the connection in the liquid line may serve as a charging valve. After evacuation and charging, the fittings are to be capped or removed. Connections shall be at least 10mm and preferably 13mm in size.

6.4.4 A permanent suction line filter may be installed in each compressor suction line. A pressure fitting must be provided ahead of the filter, preferably in the shell, to facilitate checking the pressure drop. If the pressure drop across the filter is in excess of 6.894 kPa (1 psig) after the initial 24 hours of operation, the suction line filter cartridge shall be replaced, or if the filter is of the sealed permanent type, the filter shall be replaced. Isolation block valves, by-pass piping and valve shall be provided for the suction line filter.

6.5 Air-Cooled Condensing Units

6.5.1 Neoprene pad or steel spring vibration isolators shall be provided for air cooled condensing unit. Compressors shall be mounted on a spring type vibration isolators.

6.5.2 Air-cooled condensing unit must be mounted level. Vertical air-cooled condensers shall be provided with wind deflectors.

6.5.3 Air-cooled condenser shall be strategically located to avoid hottest exposure to sun.

6.6 Drain Connections

Unless otherwise specified, condensate drains from coils and cases to the floor drain will be the responsibility of the Contractor. No drain line shall be smaller than the coil drain pan connection. All drain lines shall be hard copper tubing except for those in reach-in coolers. Lines should be sloped adequately to drain by gravity any water accumulated from condensing, defrosting, or cleaning operations. All condensate lines from refrigerated fixtures must be trapped and run to an open drain. They must not be connected directly to the sewer system. If necessary for cleaning, threaded unions shall be provided in the most accessible location near the fixture.

6.7 Factory Tests

6.7.1 All refrigerant containing parts of the air conditioning equipment shall be pressure tested in accordance with ANSI/ASHRAE Std. 15.

6.7.2 Each completely assembled refrigerant system shall be leak tested to confirm that no joint, seal or connection leaks. This leak test shall be performed prior to any painting or insulation. Leak test shall be performed in accordance with the “Factory Dehydrating, Charging and Testing” chapter of ASHRAE Handbook, Equipment Volume.

7.0 CONSTRUCTION-RELATED TESTING AND INSPECTION

7.1 General Requirements

7.1.1 The Contractor shall notify the Company at least three working days in advance of any test to be conducted so that the Company Representative and Manufacturer's Representative are present during the test.

7.1.2 Leak test, air tightness test or joint performance test of ductwork joints shall be performed while ductwork is not covered with insulation and while ceiling is not yet installed.

7.1.3 Pipe weld joints of chilled water lines (above grade and below grade) shall not be covered with insulation until radiographed and/or hydrostatic tested and passed the tests.

7.1.4 Refrigerant piping/tubing joints shall not be insulated until proven leak tight and passed pressure test.

7.1.5 Test procedures, plans, diagrams and list of equipment to be used in various tests shall be submitted by Contractor to the Company for review and approval.

7.1.6 All instruments to be used in the tests shall be calibrated and certified before and after the test.

7.1.7 All testing apparatus (hydrostatic pump, air test fan, etc.), instruments, temporary pipe work and associated works with the tests shall be provided by the Contractor.

7.1.8 10% of welded joints shall be radiographed by NDT inspector hired by the Contractor. Results of examination shall be reviewed by the Company.

7.2 Air Distribution System

7.2.1 Duct System Leakage

a) Predicted leakage rates for unsealed and sealed ducts are reviewed in ASHRAE Fundamentals Handbook. Project specification should define the allowable duct leakage, the need for leakage testing and should require the ductwork installer to perform leakage test after installing any substantial portion of the duct system.

b) Duct system constructed to 750 Pa class or lower is not recommended to be tested since it is generally recognized as not being cost effective. Close supervision and following proper installation procedures to insure air tightness should suffice to justify leakage control.

c) When tests are specified or deemed necessary, test procedures shall follow the outlines and classification in the SMACNA HVAC Duct Leakage Test Manual.

d) The allowable leakage rate shall not exceed the allotted amount for the pressure class (see Section 4 of the SMACNA Duct Leakage Test Manual) or the allotted amount for that portion of the system, whichever is applicable.

7.3 Hydronic Piping System

Chilled water and cooling water piping shall be subjected to a hydrostatic test at a minimum test pressure of 1.5 times design pressure according to procedures prescribed in ANSI B31.9; Building Services Piping.

7.4 Refrigeration System

7.4.1 Field Tests

a) After installation, purge all refrigerant lines with 7kPa dry nitrogen while soldering, brazing or welding connections. Blow out all lines with dry nitrogen prior to testing and charging.

b) After installation, all field-erected refrigerant piping, regardless of the quantity of refrigerant charge, shall be pressure-tested in accordance with the requirements of Uniform Mechanical Code (UMC) and ANSI/ASHRAE Standard 15.

Exception: Factory charged and sealed refrigerant tubing.

c) After installation, dehydrate system by placing a vacuum of 1.7 kPa absolute. Hold vacuum for 24 hours or cycle vacuum and dry nitrogen purge according to manufacturer's recommendation. Charge system with quantity and type as per manufacturer's requirements.

d) Leak testing of existing system shall be performed by injecting a mixture of a trace quantity of non-CFC refrigerant with dry nitrogen into the system. At 350 kPa, all connection shall be checked for leakage.

7.5 Radiographic Examination

10% of welded joints shall be radiographed by Non-Destructive Testing (NDT) inspector hired by the Contractor. Results of examination shall be reviewed by the Company.

8.0 PRE-START UP AND COMMISSIONING

8.1 General Requirements

All procedures and methods of documenting and verifying system performance shall be in accordance with ASHRAE Guideline for Commissioning of HVAC Systems. Specific procedures shall be submitted to NG Representative for review and approval.

8.2 Prerequisites

The following prerequisites must be met for proper start up of the system.

8.2.1 Verify completeness of physical installation of HVAC components and systems being tested.

8.2.2 Ensure all pre-start up checks have been carried out for all HVAC systems and equipment.

8.2.3 Confirm availability of permanent electrical power supply for testing operation.

8.2.4 Prior to starting the TAB process, confirm all doors, windows and ceilings are complete, doors are closed for TAB duration, area is clean and free from any influence likely to affect commissioning test measurements.

8.2.5 Ensure instrumentations and control system is available for normal operation specifically for items such as fire dampers and other motor-operated control dampers and control valves.

8.2.6 All hold tags must be released and all equipment cleared for service.

8.2.7 Electric power must be available for the operation of all motors, instruments, controls, control panels and lights.

8.2.8 Various breakers or switches must be closed to energize the equipment.

8.2.9 Lubrication must be available to all pump bearings and motors.

8.2.10 All motors must be checked for proper rotation.

8.2.11 Instrument air system must be operating and air must be available to all air-actuated controls.

8.2.12 Potable water system for make-up water must be operating, if applicable.

8.2.13 Fire protection system must be ready and operable.

8.2.14 Protective relays must be checked, adjusted and properly set.

8.2.15 Electrical equipment breakers must be certified operable.

8.2.16 Chiller compressor crankcase heaters must be energized during periods of system shutdown, if applicable.

8.3 Precautions

The following precautions must be taken before starting the system.

8.3.1 Ensure all safety checks have been carried in all equipment including power supply for controls.

8.3.2 Unusually high temperatures of either system air or chilled water must be immediately corrected.

8.3.3 Pressure safety valve on the chilled water line downstream of the pressure control valve for the input makeup water from the potable water system must be in working order.

8.3.4 With regard to the electrical equipment:

a) Follow approved procedures when working with energized electrical system.

b) Before working on electrical equipment, de-energize the system and tag all breakers open. Observe the standard precautions.

c) Before energizing the system, remove all temporary grounds, open all breakers and check that all protective relay equipment is operable.

d) Ensure that all electrical equipment is properly grounded.

8.3.5 Rotate the selection of the standby water chiller pump once a week.

8.3.6 Safety tags on equipment must be clear and legible; safety instructions must be followed exactly as given.

8.3.7 Protective covers, grounds and similar equipment must be properly installed.

8.3.8 Unusual noises in the system or equipment should be immediately investigated and corrected.

8.3.9 Do not run a pump that is dry. Purge and vent a pump before starting if the pump has been drained or if it has lost its prime.

8.3.10 Filters in the air handling units must be kept clean. The pressure drop through the pre-filter and after filter must be less than the prescribed limit. If the pressure drop is greater, clean the filters, then verify that the pressure drop is within the limit when the filters have been placed back in use.

8.3.11 The area around the system must be kept clean and free of debris.

8.3.12 The startup procedure should be followed in the order in which the steps are given to ensure the correct sequence of events and a proper startup.

8.3.13 Accurate operating records should be maintained.

9.0 TESTING, ADJUSTING AND BALANCING

9.1 General Requirements

9.1.1 After installation of the HVAC systems has successfully passed the required construction-related testing and inspection, Air/Hydronic balancing shall be scheduled and carried out by an Independent Testing Specialist Contractor not affiliated with any firm involved in design or construction phase of the project to provide an unbiased opinion of the performance of the HVAC systems.

9.1.2 All building environmental systems and components shall be checked and adjusted to produce the design objective. It shall include:

a) the balance of air and water distribution;

b) adjustment of total system to provide design quantities;

c) electrical measurements;

d) verification of performance of all equipment and automatic controls; and

e) sound and vibration measurements, when required.

9.1.3 The above tests shall be accomplished by:

a) checking installations for conformity to design requirements;

b) measurement and establishment of the fluid quantities; and

c) recording and reporting the results.

9.1.4 Prior to proceeding with the final testing activities, the following prerequisites must be completed:

a) Test procedures submitted and approved.

b) Installation complete.

c) Flushing and cleaning complete.

d) Construction-related tests (pressure, leakage, etc.) complete.

e) Quality control inspection complete.

f) Certificate of Readiness for Energizing approved.

g) Safety tagging in place.

h) Installation verification checks successfully completed.

i) Automatic controls properly functioning.

j) Applicable documents prepared and available.

9.1.5 A complete system test report shall be provided on air and water distribution systems, including flow rates, temperatures, pressures as follows:

a) Record data on air side and/or water sides of all air handling units, fans, coils, water chillers, air-cooled condensers, etc. Data shall include all air and water flow, motor, starter heaters, manufacturer, model number, type, nameplate data, etc.

b) All air moving equipment shall be balanced to within ±5% of its capacities. Balance air distribution systems through air terminals to within ±10% of air quantities shown on project drawings and record actual readings taken.

c) Adjust all chilled water balance valves for quantities to within 5% of values shown on project drawings and record actual readings.

d) Check operation of all controls to insure that all actuators cycle in accordance with design action of the controlling device. Check all control element for proper “normally open” or “normally closed” positions upon loss of power or instrument air.

e) Provide test points and plugs or covers for all opening in duct.

9.1.6 All readings shall be recorded on report forms indicating initial and final data.

9.1.7 All reports shall be reviewed and certified by a professional HVAC Engineer or TAB Supervisor or any approved equivalent and submitted for approval and acceptance.

9.1.8 Prepare notifications to the CONTRACTOR of any deficiencies which must be corrected before final balancing can be carried out to meet the specifications.

9.1.9 After final air and hydronic balancing has been successfully completed, the HVAC systems shall be left operating for at least 72 hours continuously and temperature and humidity of various key areas shall be monitored and recorded.

9.1.10 All testing procedures shall be in accordance with the procedures specified herein and industry standards applicable to any systems not covered by this specifications.

9.1.11 Tests on HVAC equipment shall be carried out at specified design outdoor conditions. However, if tests are to be conducted other than design outdoor conditions, test results shall be evaluated against the performance curves of the equipment tested. Avoid conducting equipment capacity tests below ambient temperature of 35°C.

9.2 Testing Organization

The criteria for qualifications of the Independent testing organization shall be membership in the Associated Air Balance Council (AABC) or certification by the National Environmental Balancing Bureau (NEBB) or any SMACNA accredited TAB Firm. The testing organization shall specialize in this field of work and be independent of the installing contractors or equipment suppliers for the project.

9.3 Test Procedures

Test Procedures shall be specific for any particular system and the Test and Balance Organization shall submit to National Grid SA for review and approval.

The Test and Balance Organization shall perform testing, adjusting and balancing of all equipment and systems and may use the latest general procedures published either by the following:

a) ASHRAE Standard 111, Testing, Adjusting, and Balancing of Building HVAC Systems.

b) ASHRAE - HVAC Application Handbook, "Testing, Adjusting and Balancing".

c) SMACNA "HVAC Systems - Testing, Adjusting and Balancing"

d) AABC "National Standards for Field Measurement and Instrumentation, Total System balance, Volume 1"

e) NEBB "Procedural Standard for Testing, Adjusting and Balancing of Environmental Systems"

9.3.1 General Outline for Testing Procedures

a) Air Balancing Tests

i. Obtain manufacturers certified data on all air handling equipment and terminal devices (fan curves, K-Factors, etc.).

ii. Develop a system diagram in flow sheet from with design data entered and provisions for recording field test results.

iii. Set all branch and main line dampers in full open position.

iv. Open all diffuser dampers, controls and distributions grids.

v. Check condition of belts, bearings, filters and isolators prior to start-up.

vi. Check rotation and ampere loads on all equipment.

vii. Place all equipment in continuous operation for duration of testing (including exhaust systems).

viii. Check fan speed and total static pressure related to manufacturers data for performance. Adjust pulleys as required.

ix. Adjust return air and outside air to design quantity.

x. Take an initial reading of all diffusers and enter on flow chart for analysis.

xi. Total the actual air quantities for each separate branch line (this is the sum total of diffusers on a given branch). Adjust each diffuser to deliver air quantities in relation to total.

xii. Step 10 continued for each branch duct.

xiii. Adjust each branch damper to deliver the required volume of air.

NOTE: The flow sheet recording will determine sequence of damper adjustments. Normally, flow in the branch ducts nearest the fan will be high.

xiv. Recheck total static pressure versus fan curve. Adjust pulley as required.

NOTE: Static pressure often rises as dampers are positioned.

xv. Take readings of each outlet and record on flow sheet for analysis and make final adjustments.

b) Hydronic Balancing Tests

i. Obtain manufacturers certified data on chilled water pumps (Pump performance curves, pump motor electrical data, etc.)

ii. Develop a system diagram in flow sheet form with design data entered and provisions for recording field test results.

iii. From the pressure gauges at the pump determine the total head developed by the pump. From the pump curve, determine the flow.

iv. Adjust the flow through the cooling coils by means of balancing valves to the specified quantity in liters/second.

c) Sound Level Tests

i. Measurements shall be taken at times when the building is unoccupied, or when activity in surrounding areas and background noise level in areas tested are at a minimum and relatively free from sudden changes in noise levels.

ii. Measurements shall be taken with all equipment secured, except that being tested. Tests shall be conducted only after the air balance is completed.

iii. The required sound levels shall be measured at any point within a room not less than 1.80 meters from an air terminal or room unit, and not closer than 0.90 meter from any floor, wall or ceiling surface.

iv. Sound levels shall be measured with a sound meter complying with ANSI S1.4-83, published by the American National Standards Institute. The "A" scale shall be used to measure overall sound levels. To determine the specified octave band levels, the above sound level meter, set on "C" scale, shall be supplemented by an Octave Band Analyzer complying with the latest ANSI specifications.

v. Equipment Components: Equipment components of room sound (noise) levels shall be determined for each (63 Hz. - 8 kHz.) octave bands as follows:

a. Measure room sound pressure (LP-b) with all equipment to be tested shut-off.

b. Measure room sound pressure level (LP - t) with all equipment to be tested switched on.

c. Calculate (LP-t) minus (LP-b), if this value is less than 1, applicable test must be rerun with lower background level (LP-b) unless (LP-t) is within sound pressure level specified for equipment.

d. Determine "C" from the table below:

(LP-t) - (LP-b), dB	"C"
0	10
1	7
2	4
3	3
4	2
5 to 9	1
10 or more	0

e. The "Equipment Component" of room sound level equals (LP-t) minus "C".

d) Vibration Tests

i. After installation and start-up of all rotating equipment, vibration tests shall be conducted for air handling units, ventilating fans and air-cooled condensing units, chillers, pumps, etc.

ii. Measurements shall be made using a vibration analyzer.

iii. Displacement peak to peak in mils shall be measured on each bearing radially in the X and Y axis (perpendicular to the shaft) and axially (parallel to the shaft).

iv. The following steps shall be taken to assure that equipment vibration is properly tested:

a. Determine operational speeds of the equipment from nameplates, drawings, submittals or a speed measuring device, such as tachometer or strobe, and indicate on the test form.

b. Determine acceptance criteria from specifications and indicate on the test form. If specifications do not provide criteria, use the tolerances as shown below.

c. Assure that the vibration isolation system is functioning properly.

d. Operate the equipment and make visual and audible checks for any apparent rough operation. Any defective bearing, misalignment or obvious rough operation shall be corrected before proceeding further. Where not corrected, equipment shall be considered not acceptable.

e. Measure and record on test form, vibration at bearing of driving and driven components in horizontal, vertical and, where possible, axial directions. There should be at least one axial measurement for each rotating component (fan motor, pump motor)

f. Tolerances: Vibration levels shall not exceed the following at the stated rotational speed:

Displacement

Peak-to-Peak, mm	RPM
0.140	200
0.089	300
0.064	400
0.051	500
0.046	600
0.038	700
0.036	800
0.030	900
0.025	1000
0.023	1200
0.015	1800
0.013	2000
0.009	3000
0.008	3600